

Digital Logic Assignment #7

$$1) a) \begin{array}{r} 0010 \\ + 1101 \\ \hline 1110 \end{array}$$

$$b) \begin{array}{r} 1101 \\ + 0010 \\ \hline 0011 \end{array}$$

$$c) \begin{array}{r} 0000 \\ + 1111 \\ \hline 0000 \end{array}$$

$$d) \begin{array}{r} 0001 \\ + 1110 \\ \hline 1111 \end{array}$$

$$e) \begin{array}{r} 1111 \\ + 0000 \\ \hline 0001 \end{array}$$

2)	signed	unsigned
0010	2	2
1101	-3	13
1111	-1	15
0111	7	7
1110	-2	14

$$3) \begin{array}{r} 1101 \\ + 0011 \\ \hline 0000 \end{array} \quad \begin{array}{r} 01101 \\ + 01101 \\ \hline 11010 \\ OV \end{array} \quad \begin{array}{r} 10110 \\ + 00001 \\ \hline 10111 \end{array} \quad \begin{array}{r} 1111 \\ + 0001 \\ \hline 0000 \end{array}$$

$$\begin{array}{r} 0101 \\ - 0101 \\ \hline \end{array} \quad \begin{array}{r} 1011 \\ - 0010 \\ \hline \end{array} \quad \begin{array}{r} 00101010 \\ - 00111000 \\ \hline \end{array} \quad \begin{array}{r} 1011 \\ - 0111 \\ \hline \end{array} \quad \begin{array}{r} 11000111 \\ + 1000 \\ \hline 11001000 \end{array} \quad \begin{array}{r} 1000 \\ + 1001 \\ \hline 0001 \end{array}$$

$$6) 11111011 \times 4 = 11101100$$

$$00101010 \times 4 = 10101000$$